

Spark UI, Optimizer & Internal concepts (PhysicalPlan, Jobs, Stages, Tasks)

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This document:
https://github.com/Arnaud-Nauwynck/presentations/
/pres-bigdata/12-sparkui-internals-plan-job-stage-task

Outline

What’s visible from Spark-ui

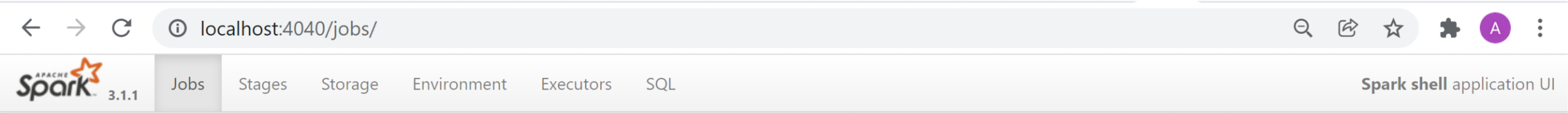
The concepts behind Dataset/RDD execution : Job, Stage, Task

The architecture of deployment : Client, Driver, Executor

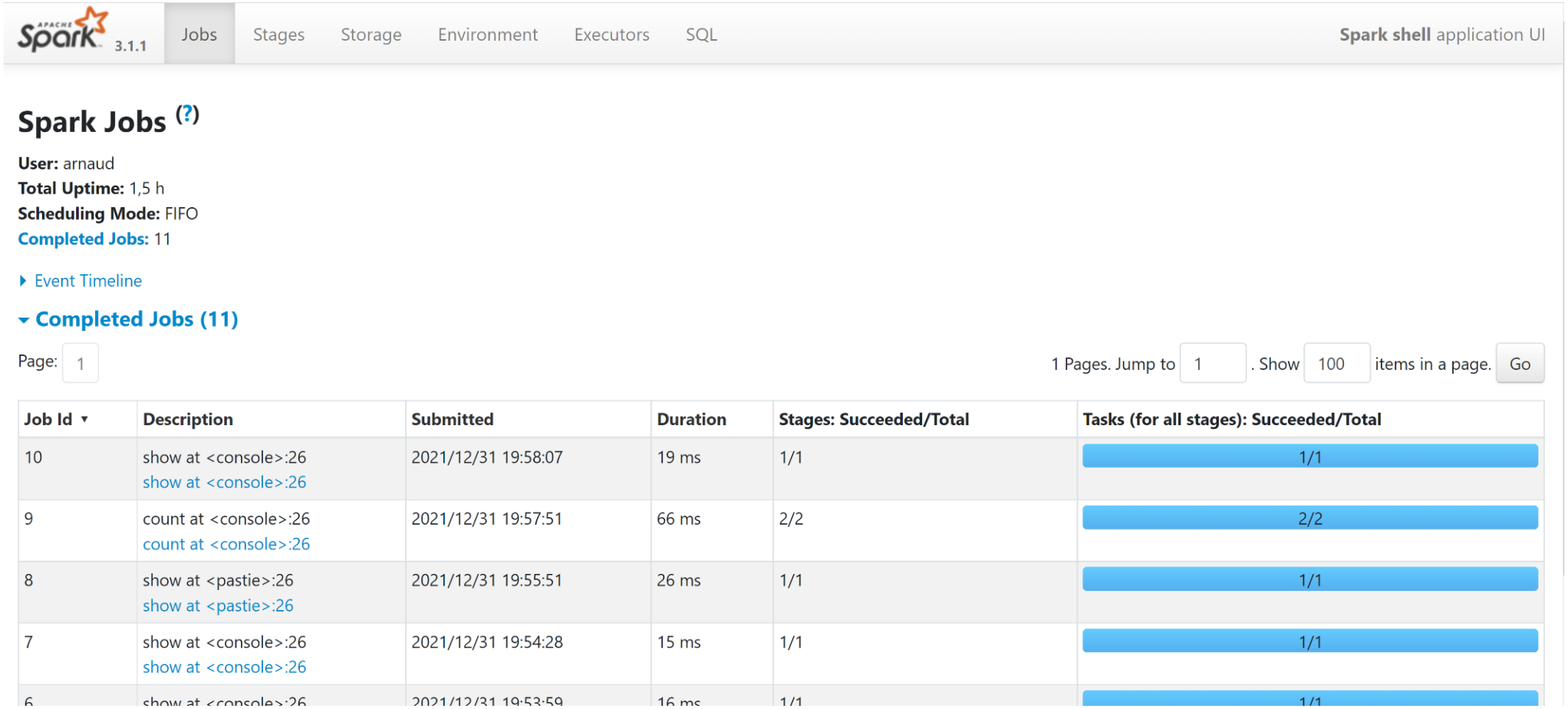
Spark-ui

```
Spark context Web UI available at http://DESKTOP-2EGCC8R:4040
Spark context available as 'sc' (master = local[*], app id = l
Spark session available as 'spark'.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 3.1.1
      /_/
```

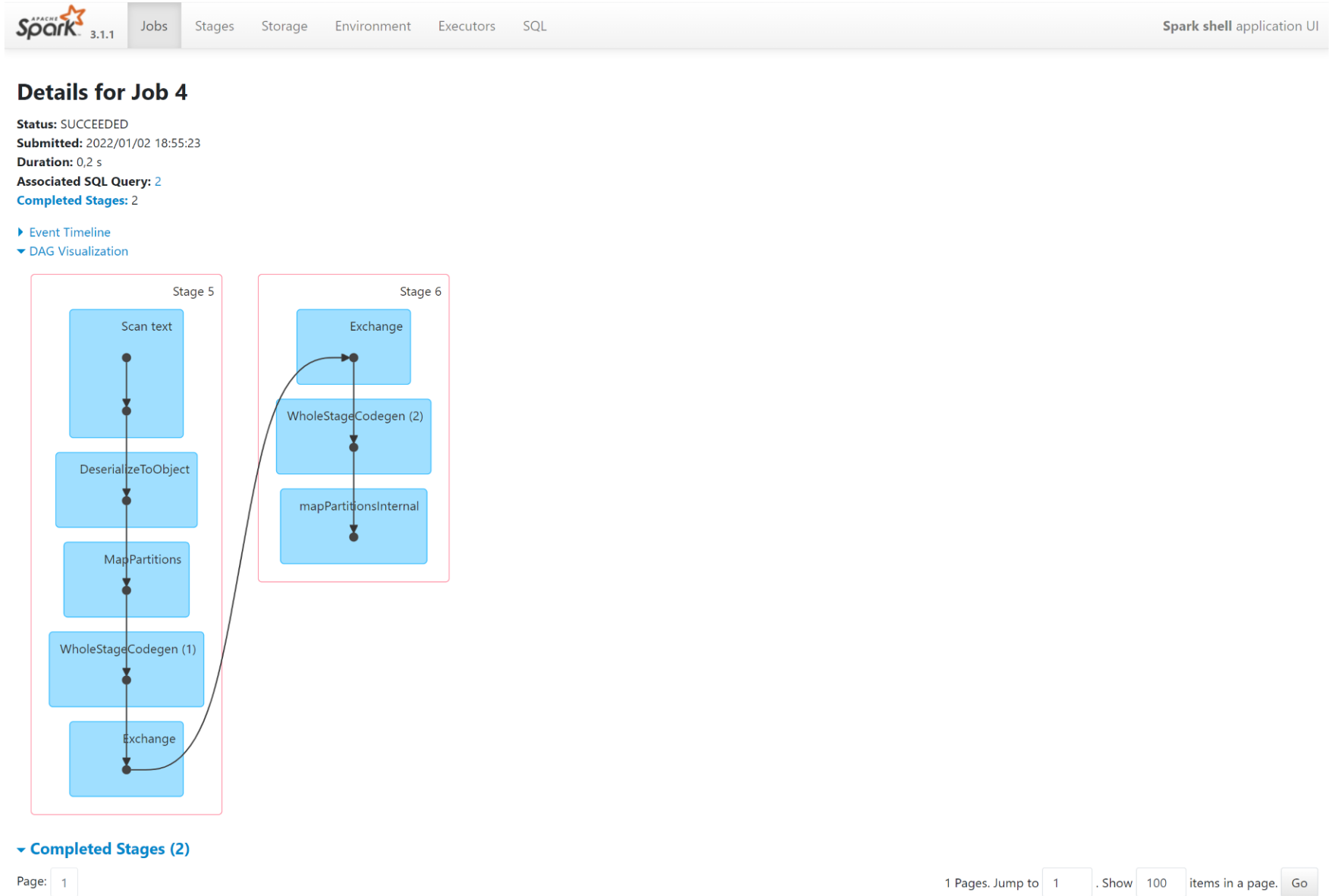
http://localhost:4040



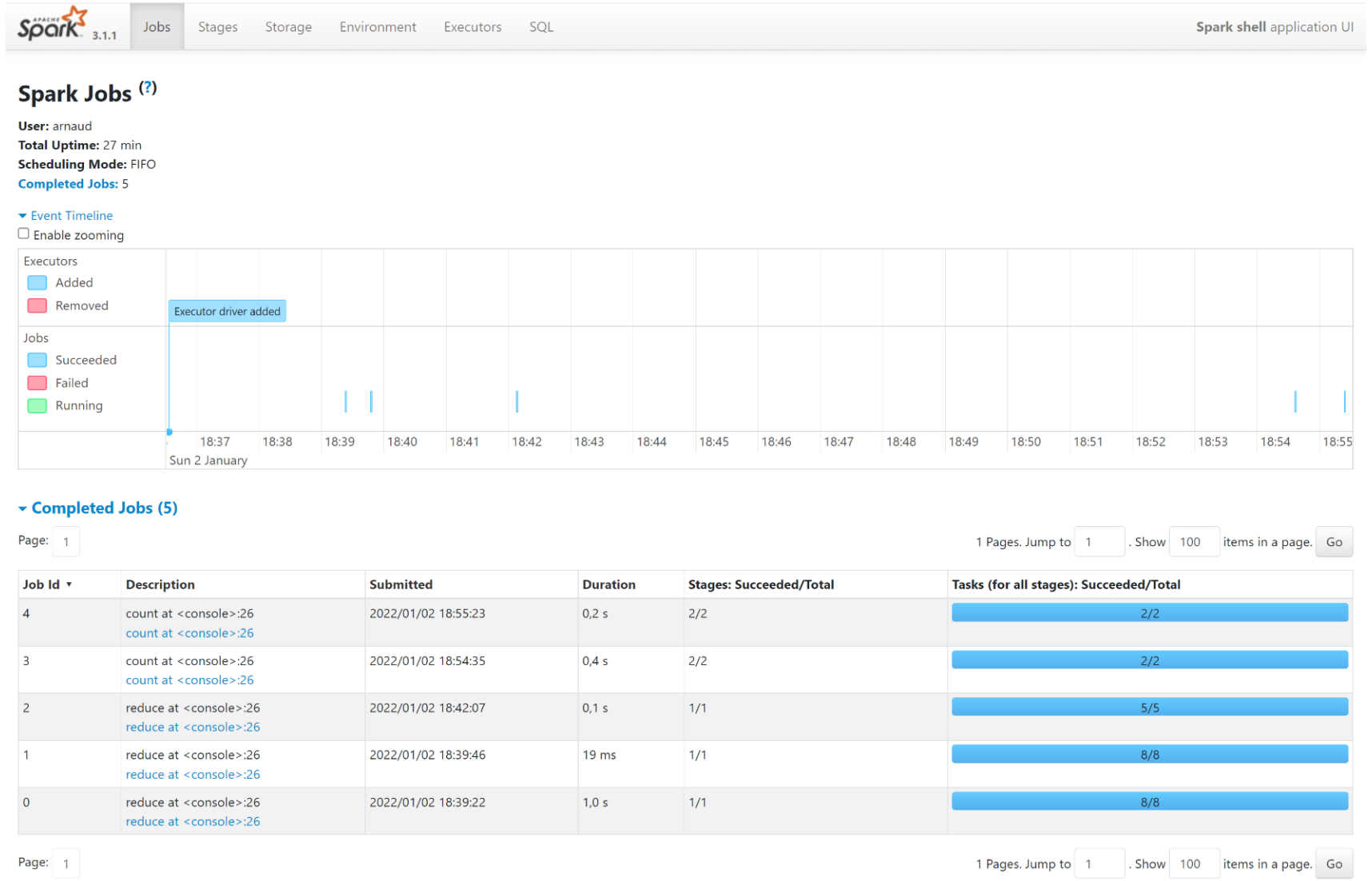
spark-ui [1/6]: Jobs



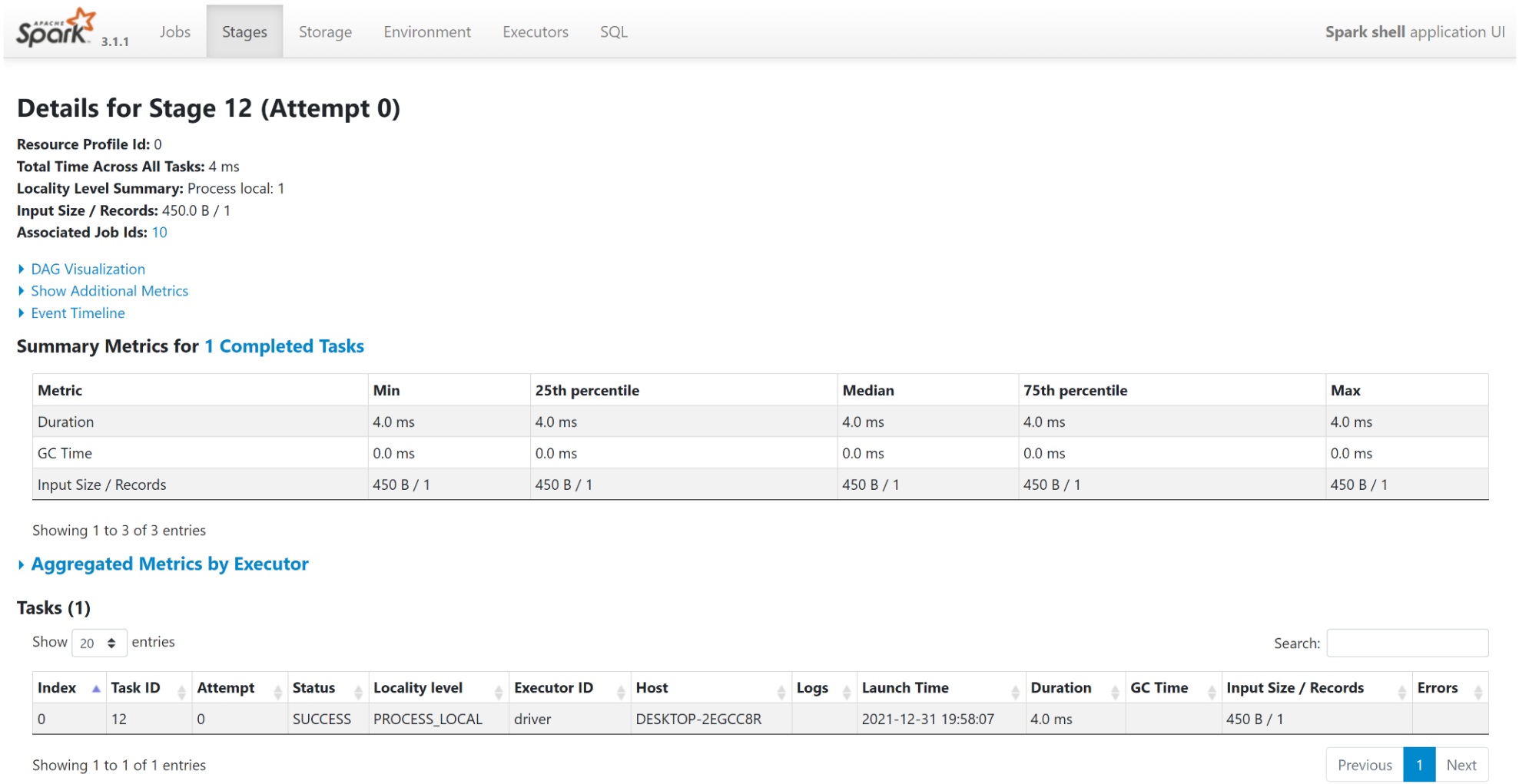
Spark-ui [1/6]: Job details DAG



Spark-ui [1/6] detail Job timeline + statistics



Spark-ui [2/6]: detail Job > Stage > Task



Spark-ui [3/6] Storage (cache, persist)



Storage

RDDs

ID	RDD Name	Storage Level	Cached Partitions	Fraction Cached	Size in Memory	Size on Disk
28	*(1) SerializeFromObject [staticinvoke(class org.apache.spark.unsafe.types.UTF8String, StringType, fromString, input[0, java.lang.String, true], true, false) AS value#30] +- MapPartitions org.apache.spark.sql.Dataset$$Lambda$4450/17435982@68a3d4, obj#29: java.lang.String +- DeserializeToObject value#21.toString, obj#28: java.lang.String +- FileScan text [value#21] Batched: false, DataFilters: [], Format: Text, Location: InMemoryFileIndex[file:/c:/data/loremIpsum.txt], PartitionFilters: [], PushedFilters: [], ReadSchema: struct<value:string>	Disk Memory Deserialized 1x Replicated	1	100%	952.0 B	0.0 B
40	FileScan text [value#21] Batched: false, DataFilters: [], Format: Text, Location: InMemoryFileIndex[file:/c:/data/loremIpsum.txt], PartitionFilters: [], PushedFilters: [], ReadSchema: struct<value:string>	Disk Memory Deserialized 1x Replicated	1	100%	960.0 B	0.0 B

Spark-ui [4/6] Environment: jvm+spark+hadoop+..

Spark 3.1.1 | Jobs | Stages | Storage | Environment | Executors | SQL | Spark shell application UI

Environment

- Runtime Information
- Spark Properties
- Resource Profiles
- Hadoop Properties
- System Properties
- Classpath Entries

Spark Properties

Name	Value
spark.app.id	local-1640975191762
spark.app.name	Spark shell
spark.app.startTime	1640975189843
spark.driver.host	DESKTOP-2EGCC8R
spark.driver.port	57122
spark.executor.id	driver
spark.home	C:\apps\hadoop\spark-3.1.1
spark.jars	
spark.master	local[*]
spark.repl.class.outputDir	C:\Users\arnaud\AppData\Local\Temp\spark-169a28ae-4479-4585-ab47-1e354d2d1347\repl-ff2cda14-38ec-4b1e-9c63-c4f02e84bf14
spark.repl.class.uri	spark://DESKTOP-2EGCC8R:57122/classes
spark.scheduler.mode	FIFO
spark.sql.catalogImplementation	hive
spark.submit.deployMode	client
spark.submit.pyFiles	
spark.ui.showConsoleProgress	true

Spark-ui [5/6]: Executors / Driver

Spark 3.1.1 | Jobs | Stages | Storage | Environment | Executors | SQL | **Spark shell** application UI

Executors

▸ Show Additional Metrics

Summary

	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Excluded
Active(1)	0	0.0 B / 413.9 MiB	0.0 B	8	0	0	13	13	0.8 s (18.0 ms)	4.6 KiB	118 B	118 B	0
Dead(0)	0	0.0 B / 0.0 B	0.0 B	0	0	0	0	0	0.0 ms (0.0 ms)	0.0 B	0.0 B	0.0 B	0
Total(1)	0	0.0 B / 413.9 MiB	0.0 B	8	0	0	13	13	0.8 s (18.0 ms)	4.6 KiB	118 B	118 B	0

Executors

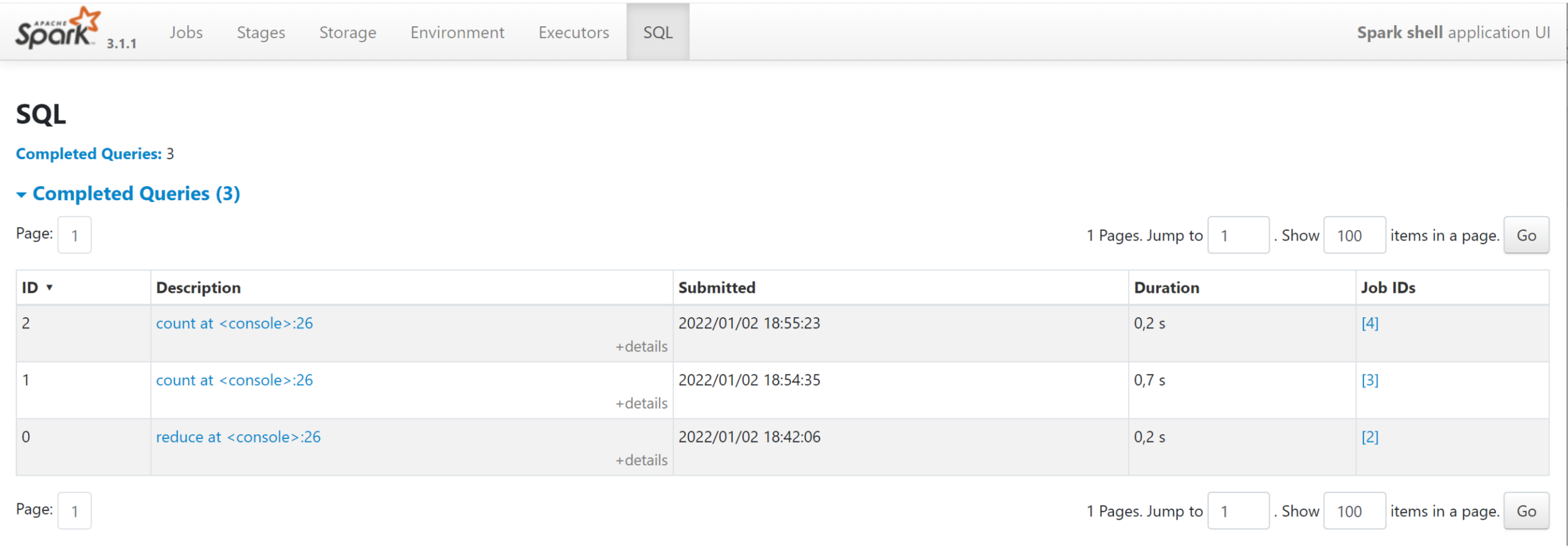
Show 20 entries Search:

Executor ID	Address	Status	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Thread Dump
driver	DESKTOP-2EGCC8R:57178	Active	0	0.0 B / 413.9 MiB	0.0 B	8	0	0	13	13	0.8 s (18.0 ms)	4.6 KiB	118 B	118 B	Thread Dump

Showing 1 to 1 of 1 entries

Previous 1 Next

Spark-ui [6/6] SQL



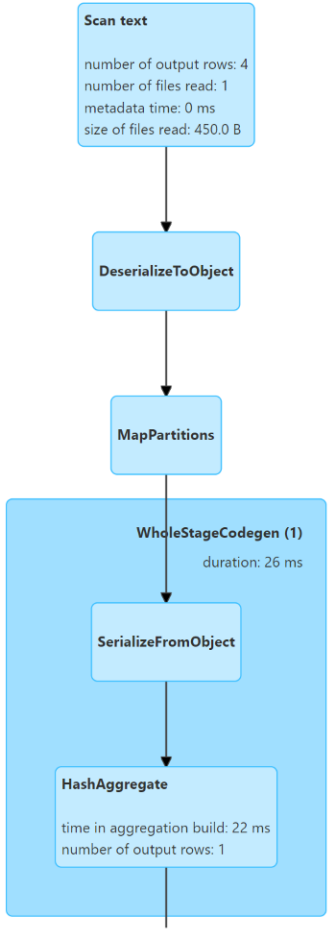
Spark-ui [6/6] Sql detail for query

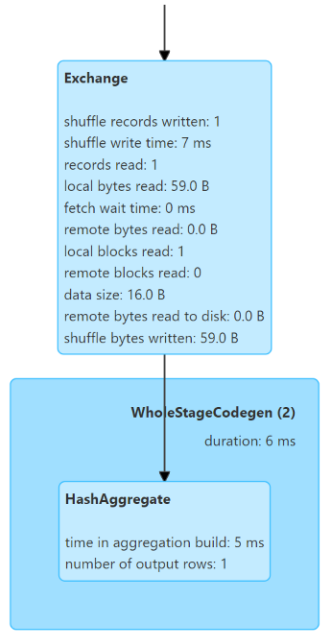
Spark 3.1.1 | Jobs | Stages | Storage | Environment | Executors | SQL

Details for Query 2

Submitted Time: 2022/01/02 18:55:23
Duration: 0,2 s
Succeeded Jobs: 4

☐ Show the Stage ID and Task ID that corresponds to the max metric





▾ Details

```
== Physical Plan ==
* HashAggregate (7)
+- Exchange (6)
   +- * HashAggregate (5)
      +- * SerializeFromObject (4)
         +- MapPartitions (3)
            +- DeserializeToObject (2)
               +- Scan text  (1)


(1) Scan text
Output [1]: [value#21]
Batched: false
Location: InMemoryFileIndex [file:/c:/data/loremIpsum.txt]
ReadSchema: struct<value:string>

(2) DeserializeToObject
Input [1]: [value#21]
Arguments: value#21.toString, obj#28: java.lang.String

(3) MapPartitions
Input [1]: [obj#28]
Arguments: org.apache.spark.sql.Dataset$$Lambda$4450/17435982@68a3d4, obj#29: java.lang.String
```

```
(4) SerializeFromObject [codegen id : 1]
Input [1]: [obj#29]

(5) HashAggregate [codegen id : 1]
Input: []
Keys: []
Functions [1]: [partial_count(1)]
Aggregate Attributes [1]: [count#35L]
Results [1]: [count#36L]

(6) Exchange
Input [1]: [count#36L]
Arguments: SinglePartition, ENSURE_REQUIREMENTS, [id=#55]

(7) HashAggregate [codegen id : 2]
Input [1]: [count#36L]
Keys: []
Functions [1]: [count(1)]
Aggregate Attributes [1]: [count(1)#32L]
Results [1]: [count(1)#32L AS count#33L]
```

Spark Internals

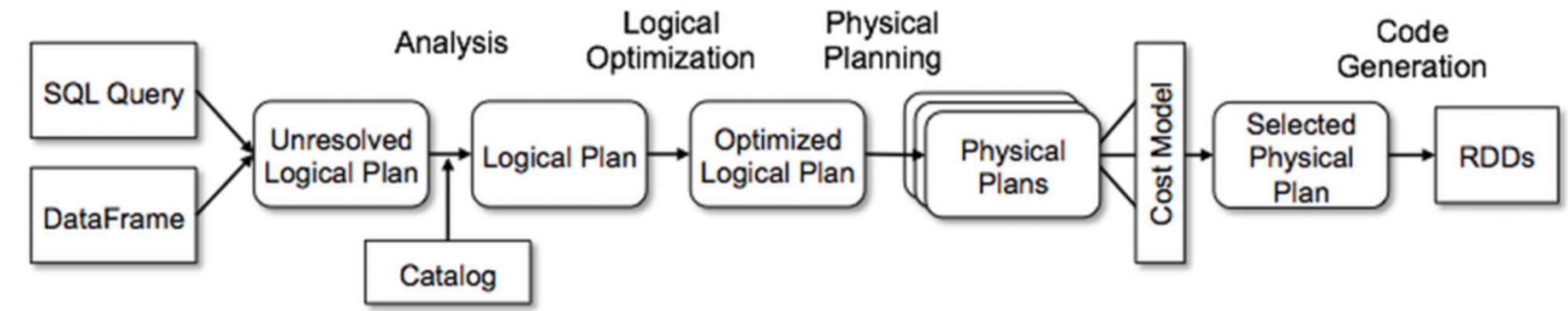
Logical/PhysicalPlan

Jobs- Stage- Tasks

Sql-> parsed AST-> Dataset + expression API

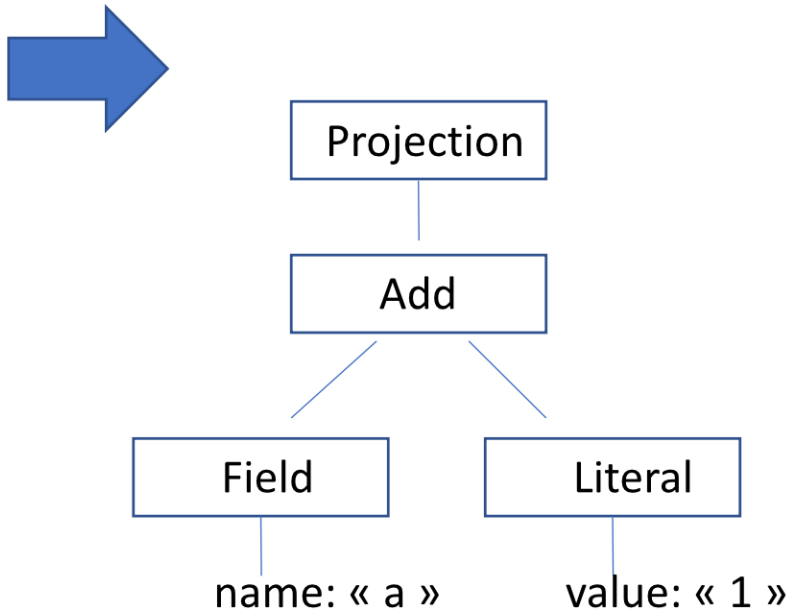
Optimisation: re-write
from LogicalPlan to PhysicalPlan

SQL / DataSet-> AST-> Catalyst-> Code + RDD



Compiler chain: Parser-> AST-> ..

Sql> select **(a+1)** from …

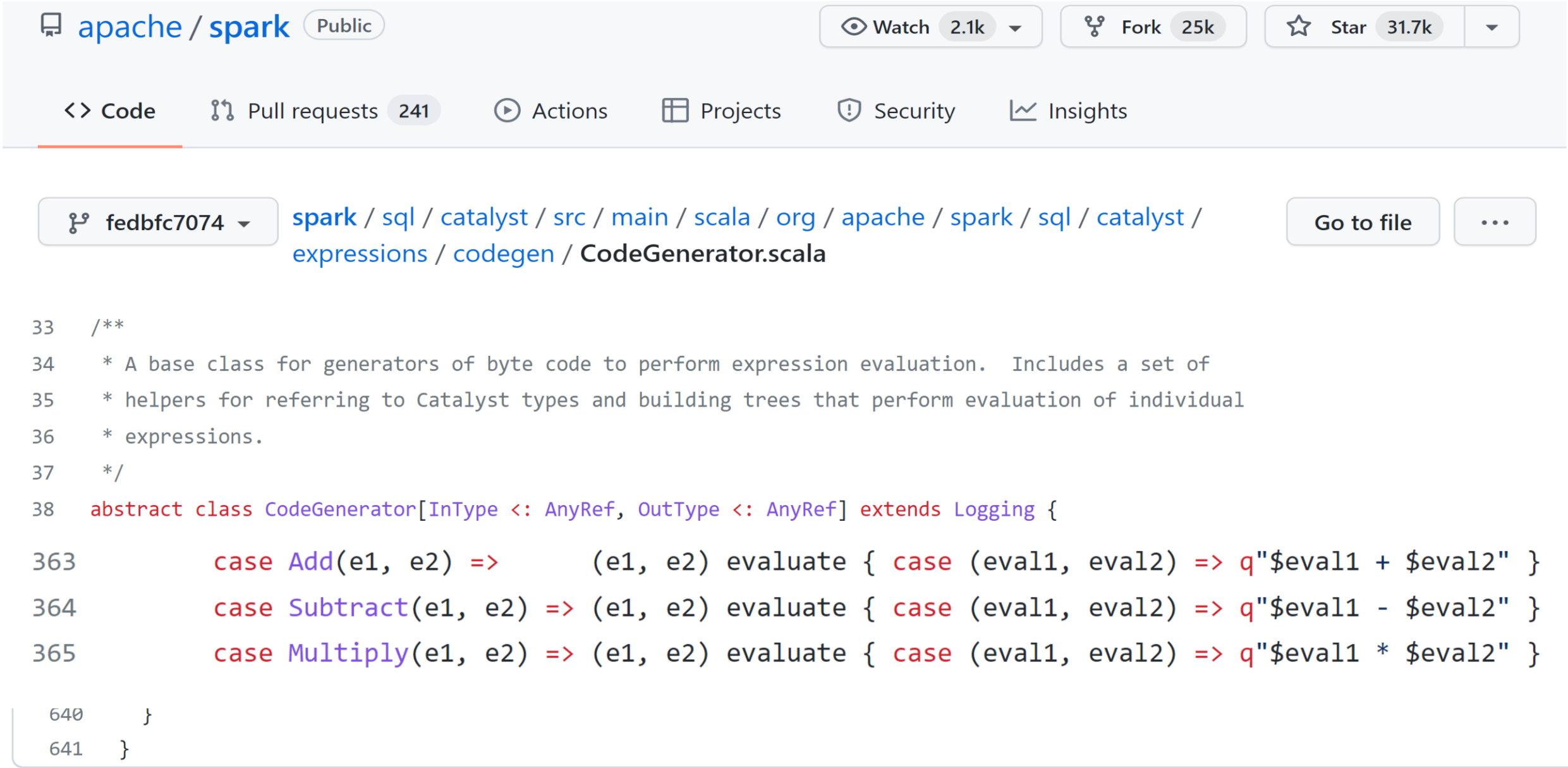



```scala
case class Add(left: Expression, right: Expression) extends BinaryArithmetic {
  override def symbol: String = "+"

  lazy val numeric = dataType match {
    case n: NumericType => n.numeric.asInstanceOf[Numeric[Any]]
    case other => sys.error(s"Type $other does not support numeric operations")
  }

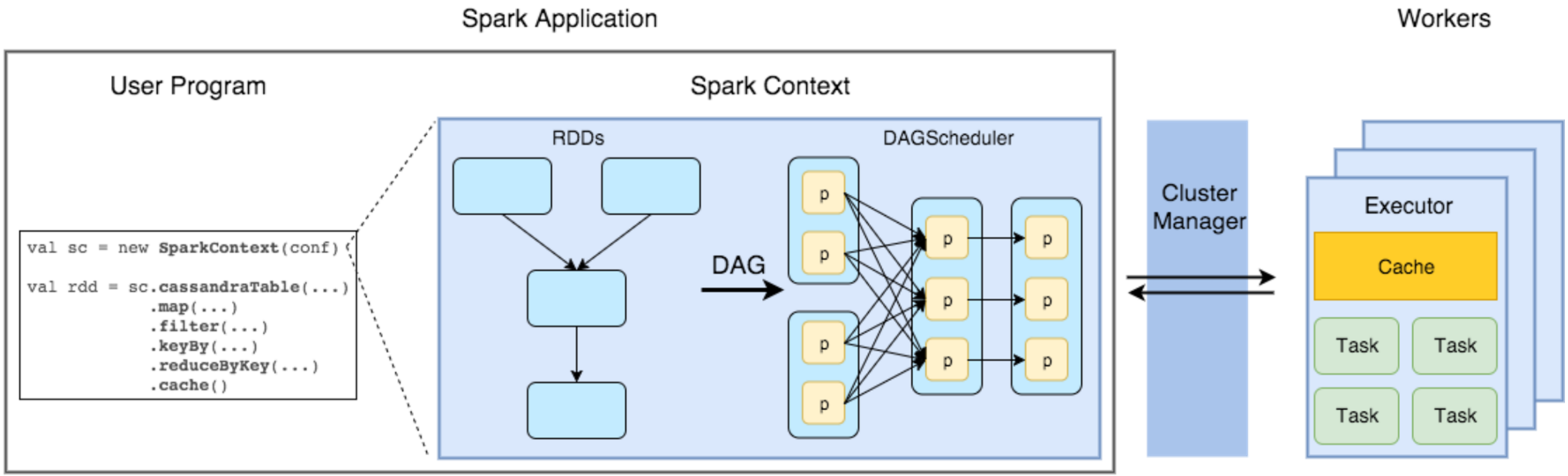
  override def eval(input: Row): Any = {
    val evalE1 = left.eval(input)
    if(evalE1 == null) {
      null
    } else {
      val evalE2 = right.eval(input)
      if (evalE2 == null) {
        null
      } else {
        numeric.plus(evalE1, evalE2)
      }
    }
  }
}
```

Catalyst AST-> CodeGenerator



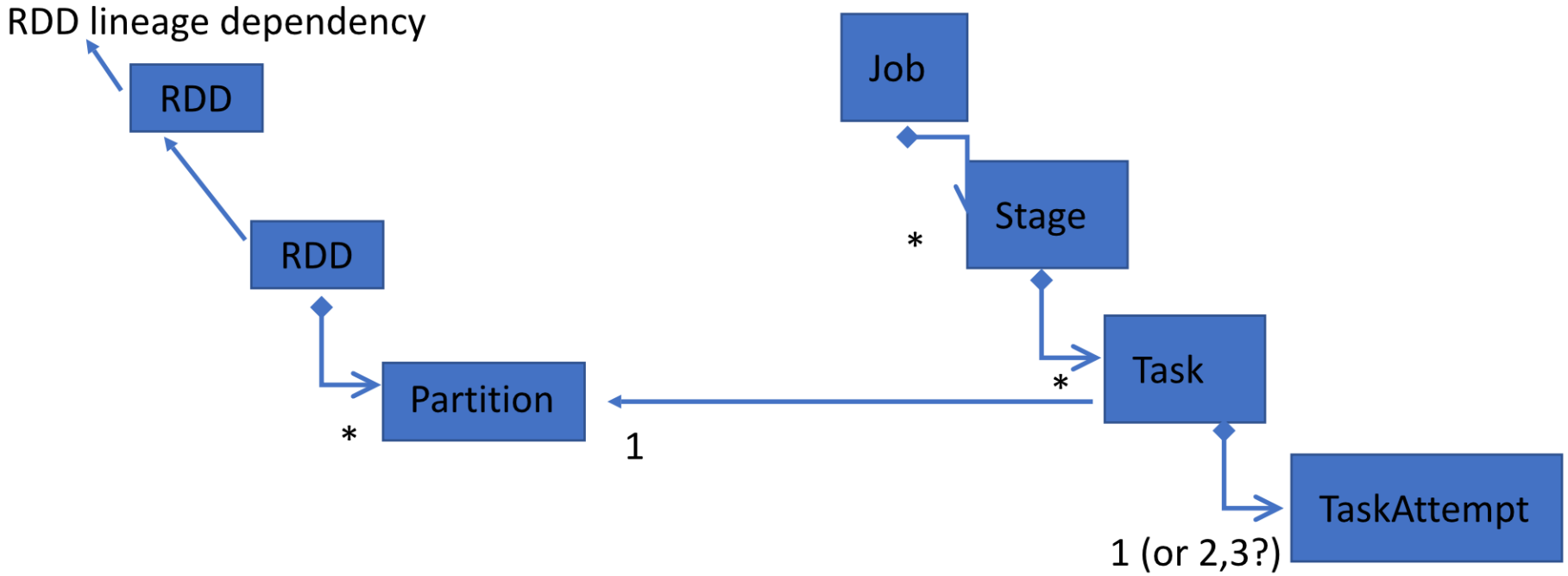
Runtime Execution:

Dataset-> RDD-> Dag Scheduler
-> Jobs,Stages,Tasks

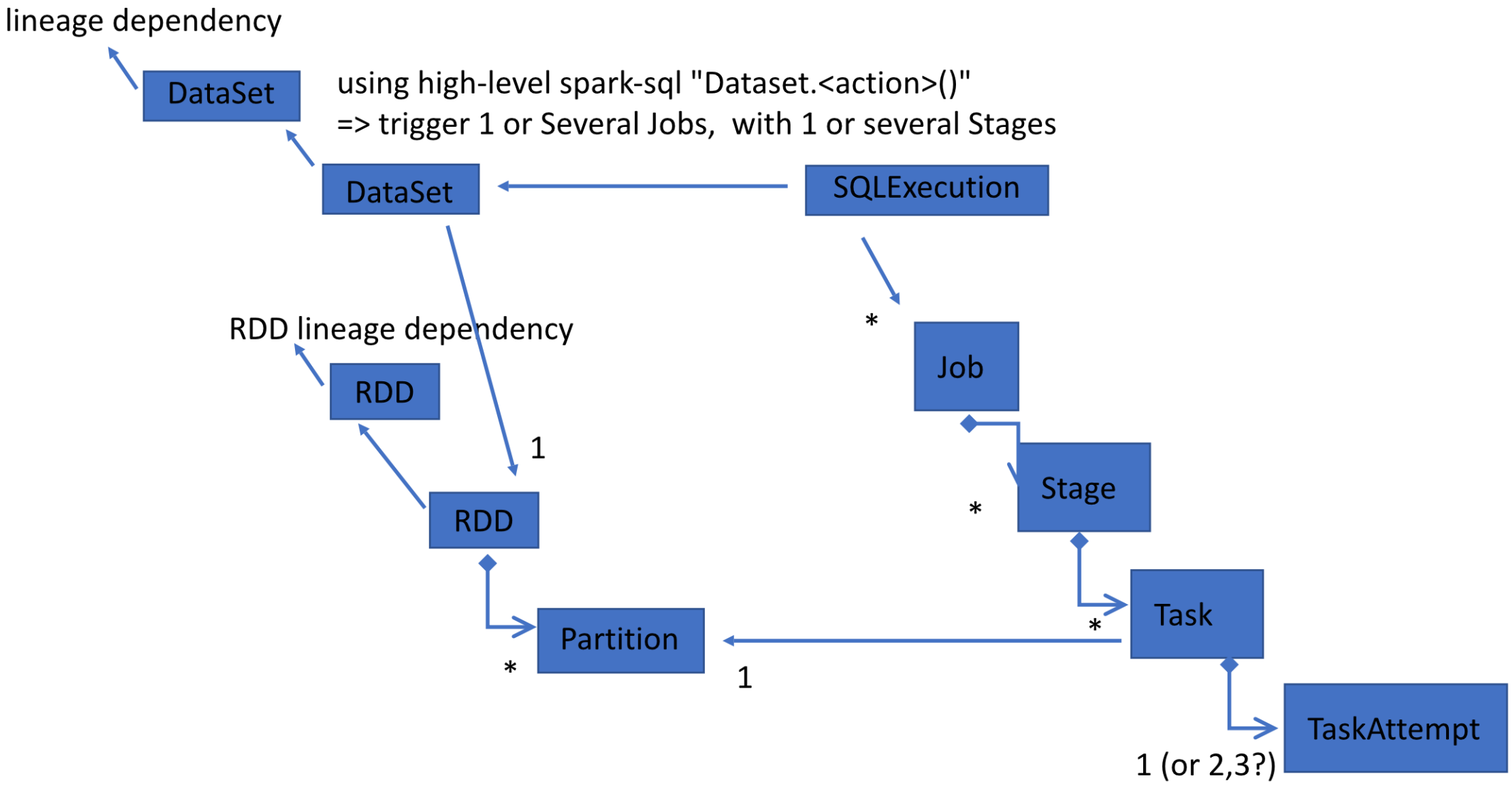

Spark Application
Workers
User Program
Spark Context
RDDs
DAGScheduler
DAG
val sc = new SparkContext(conf)
val rdd = sc.cassandraTable(...)
.map(...)
.filter(...)
.keyBy(...)
.reduceByKey(...)
.cache()
p
Cluster Manager
Executor
Cache
Task
Task
Task
Task

Spark-core Internal Concepts…
RDD, Job, Stage, Task

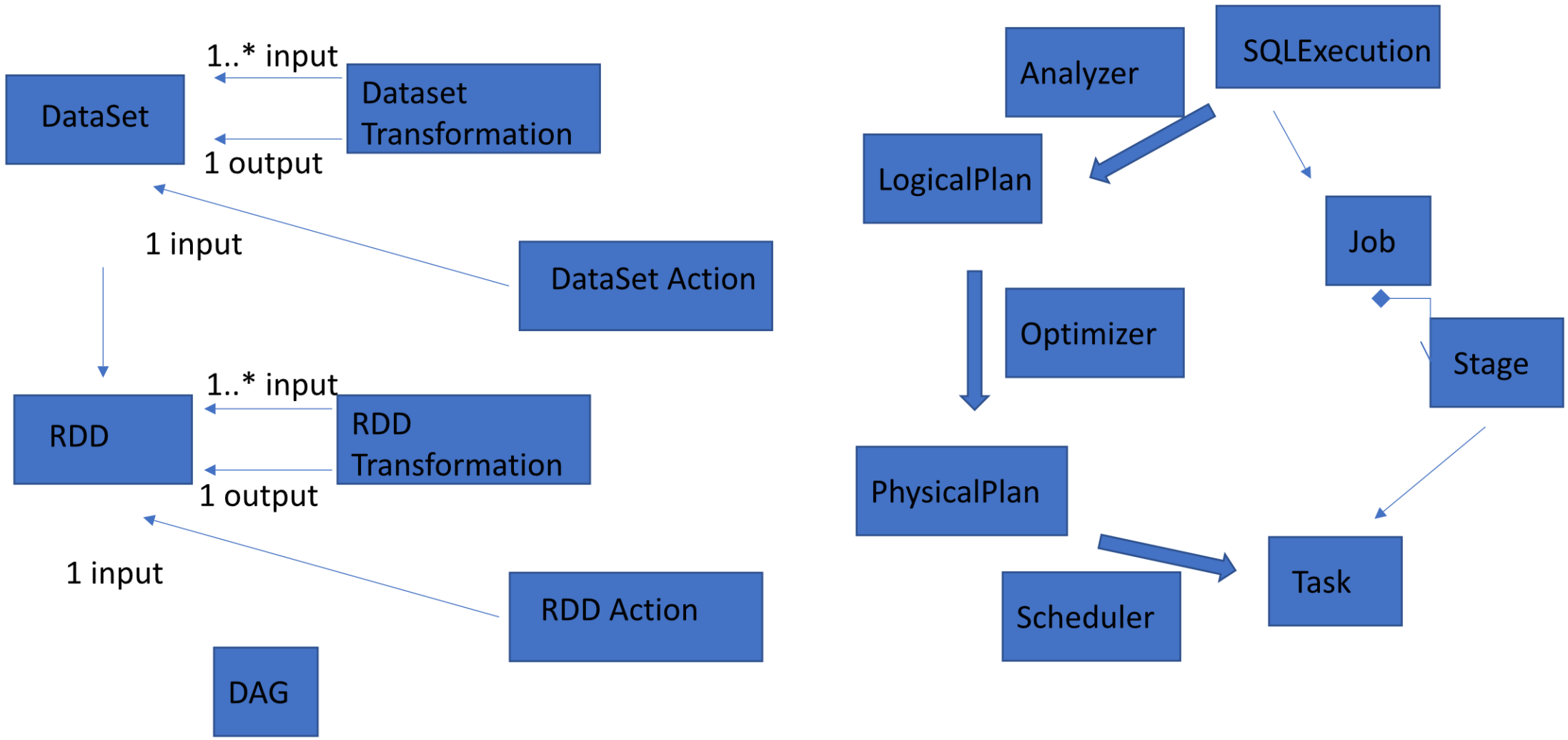
using only low-level spark-core "RDD.eval()"
=> trigger 1 Job, with 1 or several Stages



Spark-sql Internal Concepts… SQLExecution, Job, Stage, Task



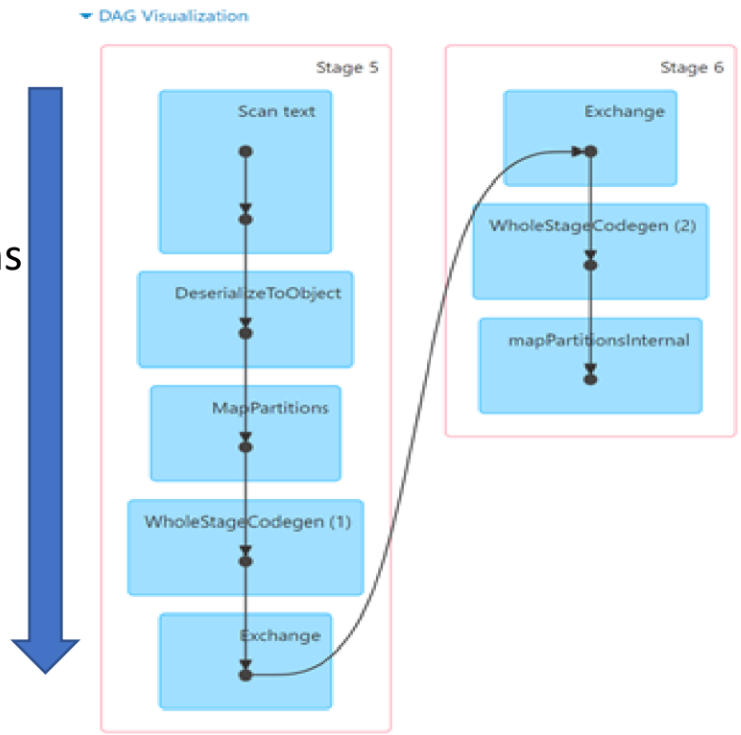
Spark Internal Concepts…
SQL, Job, Stage, Task



DAG Job = list of Stages
each Stage = N "narrow" transformations
between Stages = Exchange Shuffle ("Wide" transfo.)

1 stage =
Successive **Narrow** transformations

Bytecode generated
… executed within same
Thread/Process



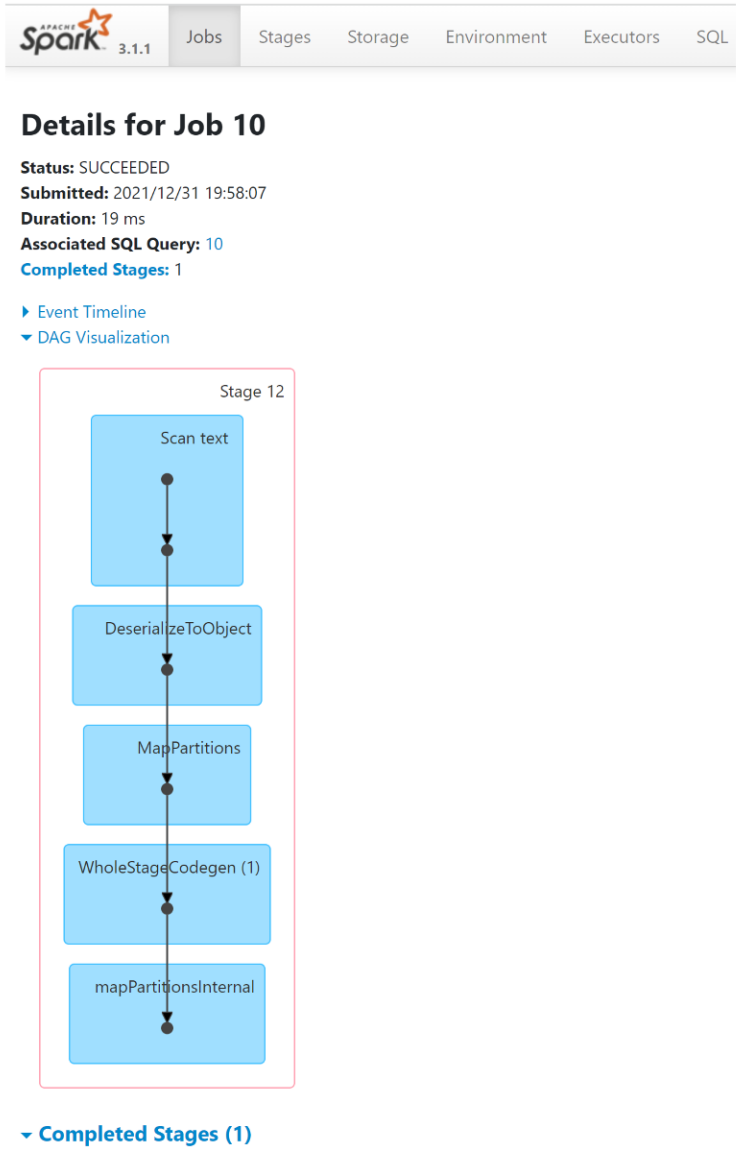
Job =
Full DAG of Stages

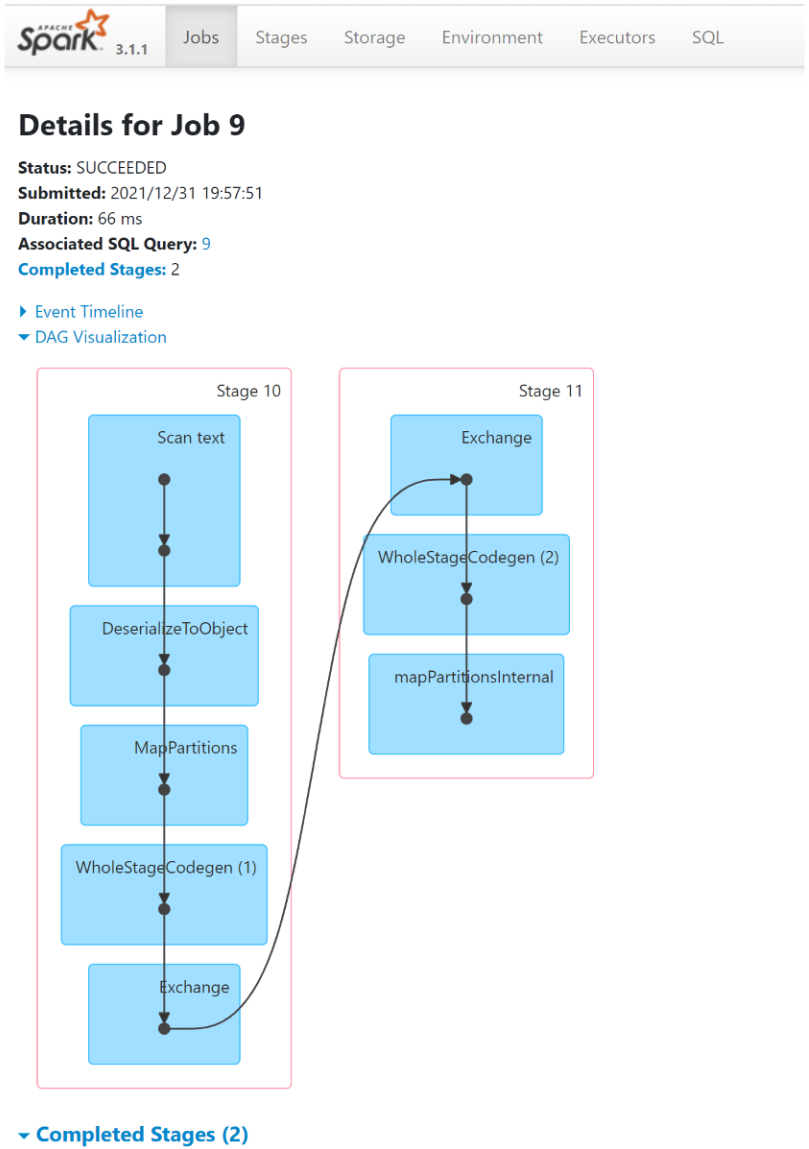
Exchange
Output

Exchange
Input

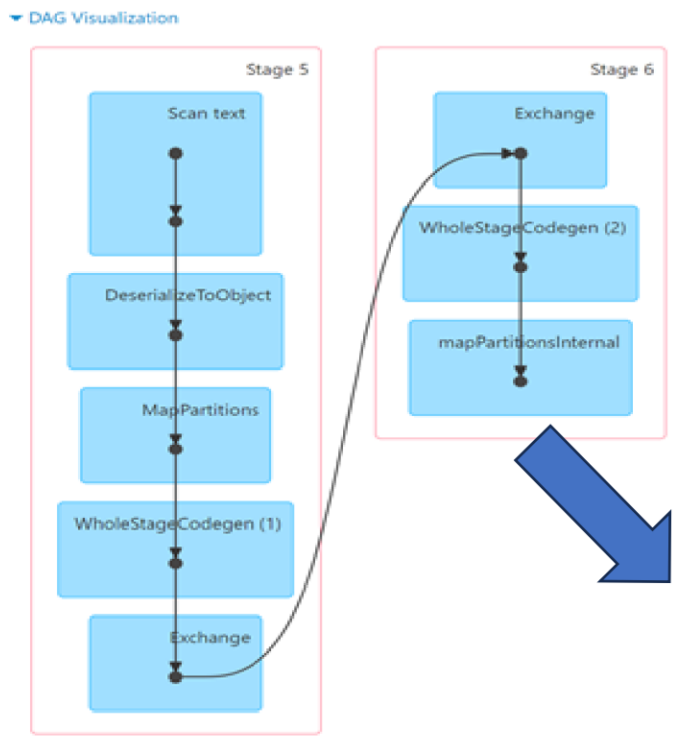
Wide transformation … separation between successive **Stages**

example Word Count DAGs simple-vs- repartition()

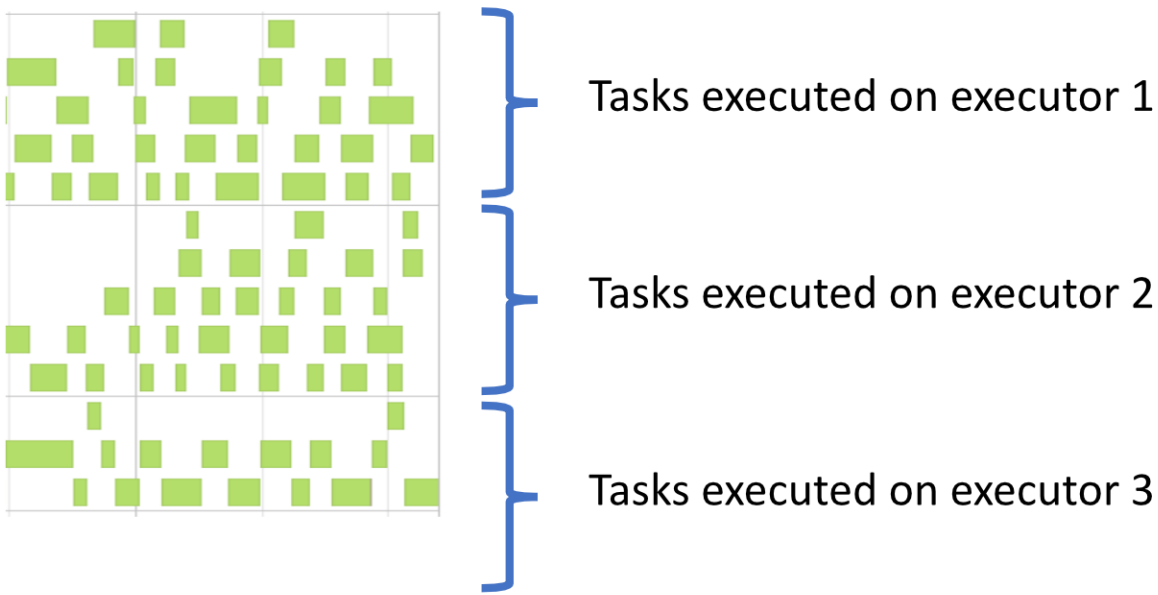




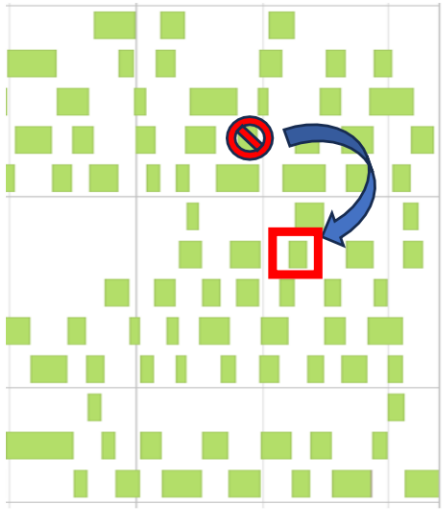
Stage = list of Tasks : 1 per Partition
... each scheduled on an executor



Tasks "**eventTimeline**" for stage



Tasks may fail, Retry individually

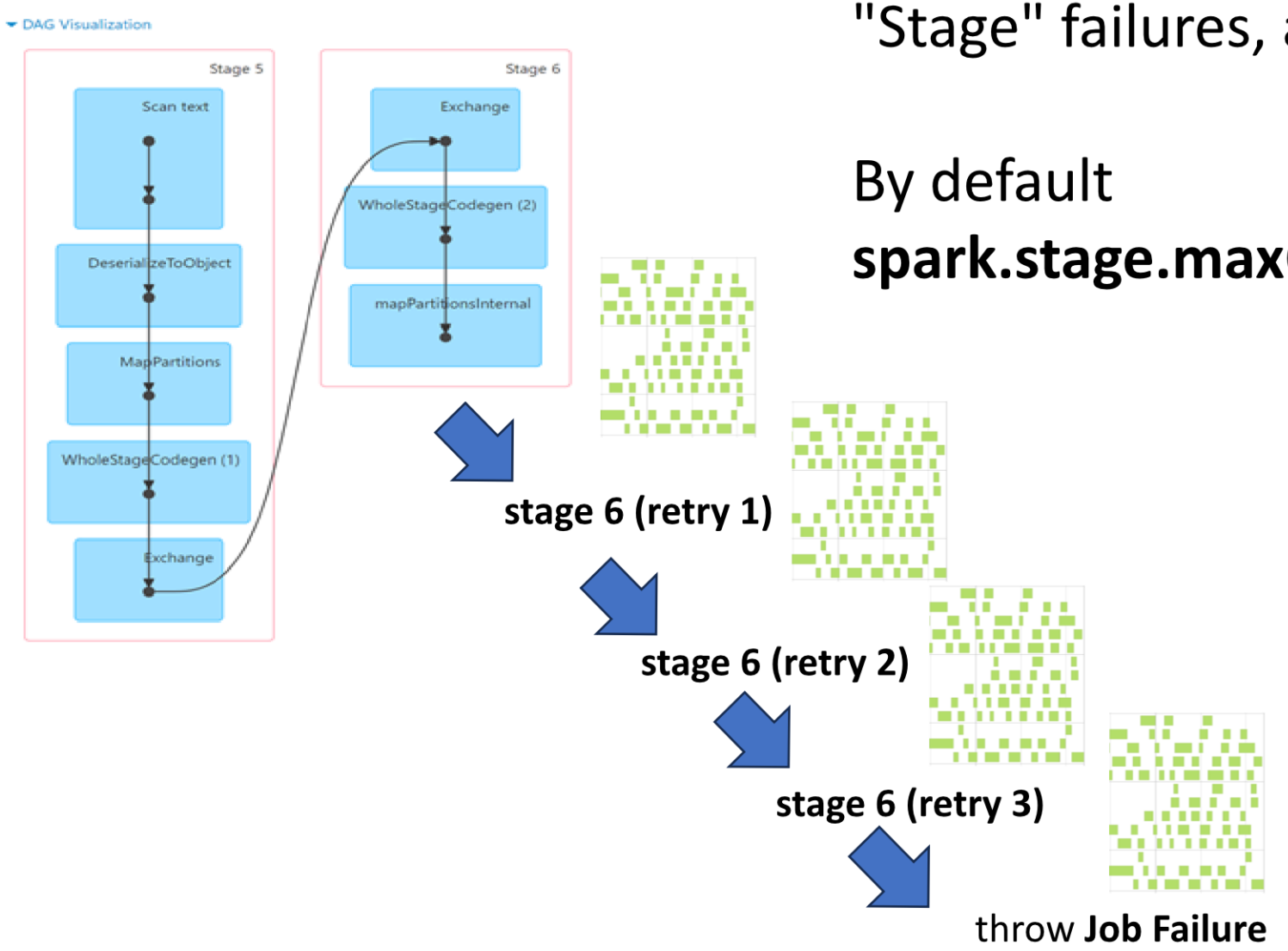


Tasks errors are automatically retried (rescheduled)

By default,

spark.task.maxFailures = 4

Stage Retry



"Stage" failures, are also retried

By default

spark.stage.maxConsecutiveAttempts = 4

Spark History Server

How to analyze a finished spark-submit App?

Typical use-case :

most daily batches run once per day, late or during night

How to analyze performances, but OFFLINE ?
(process finished => not able to connect to « http://localhost:4040 »)

How to compare performances with older applications

eventLog Files, disabled by default

Extract of file ${SPARK_HOME} / conf / spark-defaults.conf.template

```
# spark.eventLog.enabled     true
# spark.eventLog.dir         hdfs://namenode:8021/directory
…
```

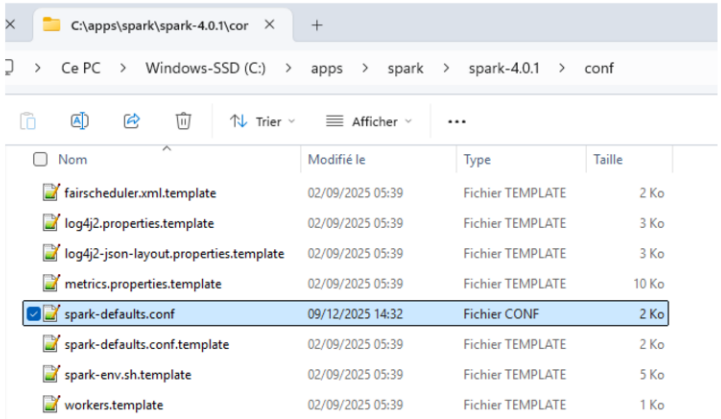
By default, eventLog Files are disabled

(no « spark-defaults.conf » file, and in the template it is commented « # »)

Notice: eventLog Files are DIFFERENT FROM   log4j stdout files
By default, there is also NO log file (only WARN message are written to console)

Enabling eventLog Files

Copy file spark-defaults.conf.template to spark-defaults.conf



+ Edit to uncomment and configure log dir

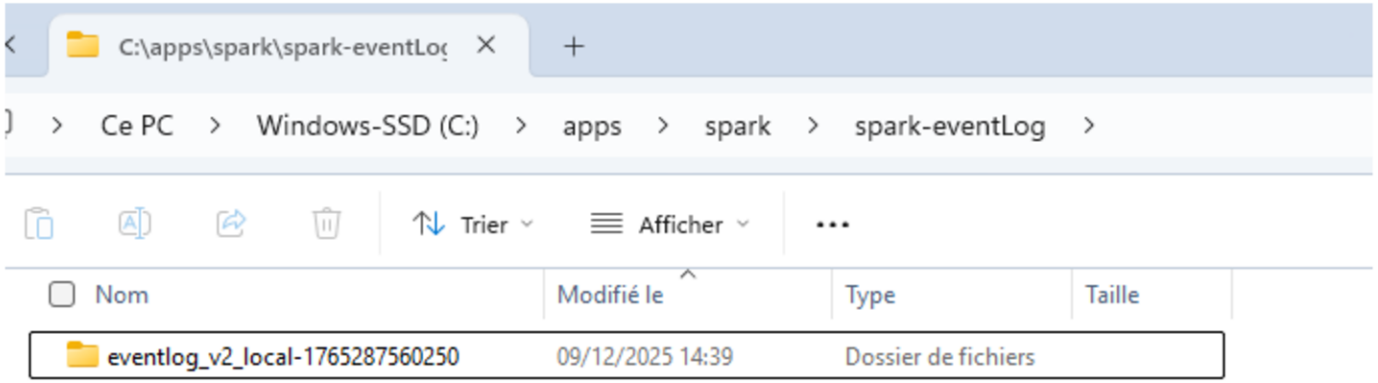
spark.eventLog.enabled true
spark.eventLog.dir file:///C:/apps/spark/spark-eventLog

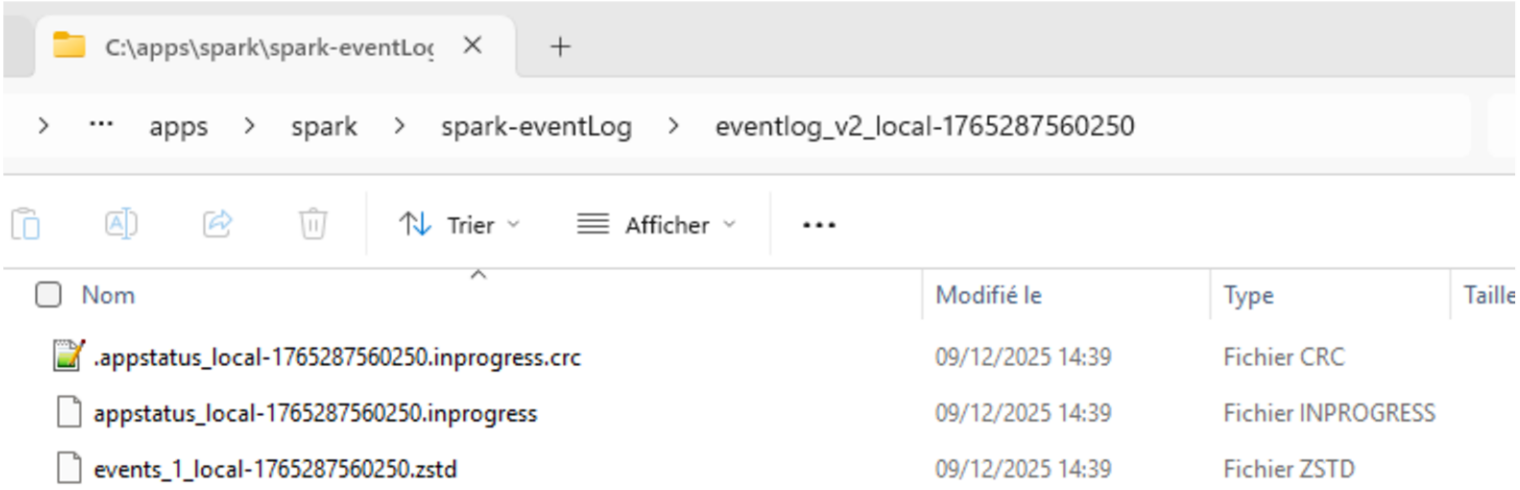
also edit for HistoryServer
spark.history.fs.logDirectory file:///C:/apps/spark/spark-eventLog

Or also explicit launch:

```
spark-shell \
    --conf spark.eventLog.enabled=true  \
    --conf spark.eventLog.dir=file:///C:/apps/spark/spark-eventLog
```

Launching spark => 1 Dir + N x written Files





Uncompressing EventLog Files, internal Check

```
arnaud@DesktopArnaud /cygdrive/c/apps/spark/spark-eventLog/eventlog_v2_local-1765287560250
$ ls -l
total 96
-rw-rw----+ 1 arnaud Aucun     0 Dec  9 14:39 appstatus_local-1765287560250.inprogress
-rw-rw----+ 1 arnaud Aucun 64997 Dec  9 14:42 events_1_local-1765287560250.zstd

arnaud@DesktopArnaud /cygdrive/c/apps/spark/spark-eventLog/eventlog_v2_local-1765287560250
$ cat events_1_local-1765287560250.zstd | zstd -d > events_1_local-1765287560250.ndjson

arnaud@DesktopArnaud /cygdrive/c/apps/spark/spark-eventLog/eventlog_v2_local-1765287560250
$ ls -l
total 836
-rw-rw----+ 1 arnaud Aucun      0 Dec  9 14:39 appstatus_local-1765287560250.inprogress
-rw-r--r--+ 1 arnaud Aucun 753747 Dec  9 14:45 events_1_local-1765287560250.ndjson
-rw-rw----+ 1 arnaud Aucun  64997 Dec  9 14:42 events_1_local-1765287560250.zstd

arnaud@DesktopArnaud /cygdrive/c/apps/spark/spark-eventLog/eventlog_v2_local-1765287560250
$ less events_1_local-1765287560250.ndjson

arnaud@DesktopArnaud /cygdrive/c/apps/spark/spark-eventLog/eventlog_v2_local-1765287560250
$ head -4 events_1_local-1765287560250.ndjson
{"Event":"SparkListenerLogStart","Spark Version":"4.0.1"}
{"Event":"SparkListenerResourceProfileAdded","Resource Profile Id":0,"Executor Resource Requests":{"cores":{"Resource Name":"cores","Amount":1,"Discovery
 Script":"","Vendor":""},"memory":{"Resource Name":"memory","Amount":1024,"Discovery Script":"","Vendor":""},"offHeap":{"Resource Name":"offHeap","Amount
":0,"Discovery Script":"","Vendor":""}},"Task Resource Requests":{"cpus":{"Resource Name":"cpus","Amount":1.0}}}
{"Event":"SparkListenerExecutorAdded","Timestamp":1765287560391,"Executor ID":"driver","Executor Info":{"Host":"DesktopArnaud","Total Cores":8,"Log Urls"
:{},"Attributes":{},"Resources":{},"Resource Profile Id":0}}
{"Event":"SparkListenerBlockManagerAdded","Block Manager ID":{"Executor ID":"driver","Host":"DesktopArnaud","Port":64835},"Maximum Memory":3032481792,"Ti
mestamp":1765287560476,"Maximum Onheap Memory":3032481792,"Maximum Offheap Memory":0}
```

eventLog Files Source Details…

eventLog = ND-JSON file containing internal SparkListener Bus events

https://github.com/apache/spark/blob/master/core/src/main/scala/org/apache/spark/scheduler/SparkListener.scala

spark / core / src / main / scala / org / apache / spark / scheduler / SparkListener.scala

Code | Blame | 588 lines (476 loc) · 18.4 KB

```scala
package org.apache.spark.scheduler

import java.util.Properties
import javax.annotation.Nullable

import scala.collection.Map

import org.apache.spark.TaskEndReason
import org.apache.spark.annotation.{DeveloperApi, Since}
import org.apache.spark.executor.{ExecutorMetrics, TaskMetrics}
import org.apache.spark.resource.ResourceProfile
import org.apache.spark.scheduler.cluster.ExecutorInfo
import org.apache.spark.storage.{BlockManagerId, BlockUpdatedInfo}

@DeveloperApi
case class SparkListenerStageSubmitted(stageInfo: StageInfo, properties: Properties = null)
  extends SparkListenerEvent

@DeveloperApi
case class SparkListenerStageCompleted(stageInfo: StageInfo) extends SparkListenerEvent

@DeveloperApi
case class SparkListenerTaskStart(stageId: Int, stageAttemptId: Int, taskInfo: TaskInfo)
  extends SparkListenerEvent

@DeveloperApi
case class SparkListenerTaskGettingResult(taskInfo: TaskInfo) extends SparkListenerEvent

@DeveloperApi
case class SparkListenerSpeculativeTaskSubmitted(
    stageId: Int,
    stageAttemptId: Int = 0)
```

https://github.com/apache/spark/blob/master/core/src/main/scala/org/apache/spark/util/JsonProtocol.scala

spark / core / src / main / scala / org / apache / spark / util / JsonProtocol.scala ↑ Top

Code | Blame | 1705 lines (1568 loc) · 75.4 KB | Raw

```scala
  val defaultOptions: JsonProtocolOptions = new JsonProtocolOptions(new SparkConf(false))
  def sparkEventToJsonString(event: SparkListenerEvent, options: JsonProtocolOptions): String = {
  }

def writeSparkEventToJson(
    event: SparkListenerEvent,
    g: JsonGenerator,
    options: JsonProtocolOptions): Unit = {
  event match {
    case stageSubmitted: SparkListenerStageSubmitted =>
      stageSubmittedToJson(stageSubmitted, g, options)
    case stageCompleted: SparkListenerStageCompleted =>
      stageCompletedToJson(stageCompleted, g, options)
    case taskStart: SparkListenerTaskStart =>
      taskStartToJson(taskStart, g, options)
    case taskGettingResult: SparkListenerTaskGettingResult =>
      taskGettingResultToJson(taskGettingResult, g, options)
    case taskEnd: SparkListenerTaskEnd =>
      taskEndToJson(taskEnd, g, options)
    case jobStart: SparkListenerJobStart =>
      jobStartToJson(jobStart, g, options)
    case jobEnd: SparkListenerJobEnd =>
      jobEndToJson(jobEnd, g)
    case environmentUpdate: SparkListenerEnvironmentUpdate =>
      environmentUpdateToJson(environmentUpdate, g)
    case blockManagerAdded: SparkListenerBlockManagerAdded =>
      blockManagerAddedToJson(blockManagerAdded, g)
    case blockManagerRemoved: SparkListenerBlockManagerRemoved =>
      blockManagerRemovedToJson(blockManagerRemoved, g)
    case unpersistRDD: SparkListenerUnpersistRDD =>
      unpersistRDDToJson(unpersistRDD, g)
    case applicationStart: SparkListenerApplicationStart =>
```

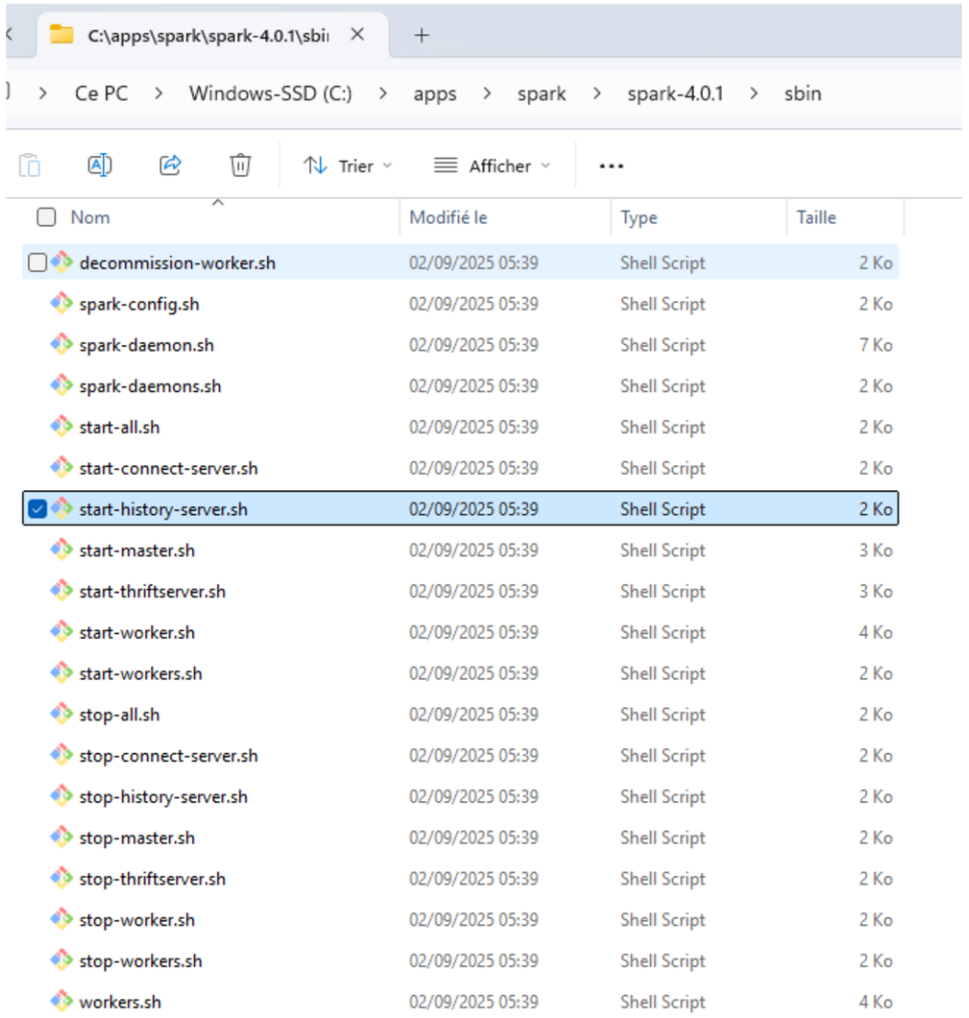
Running Spark History Server

The History Server is the same code that Spark UI,
but running as a Web Server to view old eventLog Files

To run it …

${SPARK_HOME}/sbin/start-history-server.sh

Notice:
- in « sbin » directory, not in « bin » (so not in your PATH)
- Only « .sh » bash script, no windows « .cmd »



Spark History Server …

sbin/spark-history-server.sh

➔ does not work on windows, even with cmder + bash

➔ is almost equivalent to …

spark-class.cmd org.apache.spark.deploy.history.HistoryServer

➔ start but expect different conf

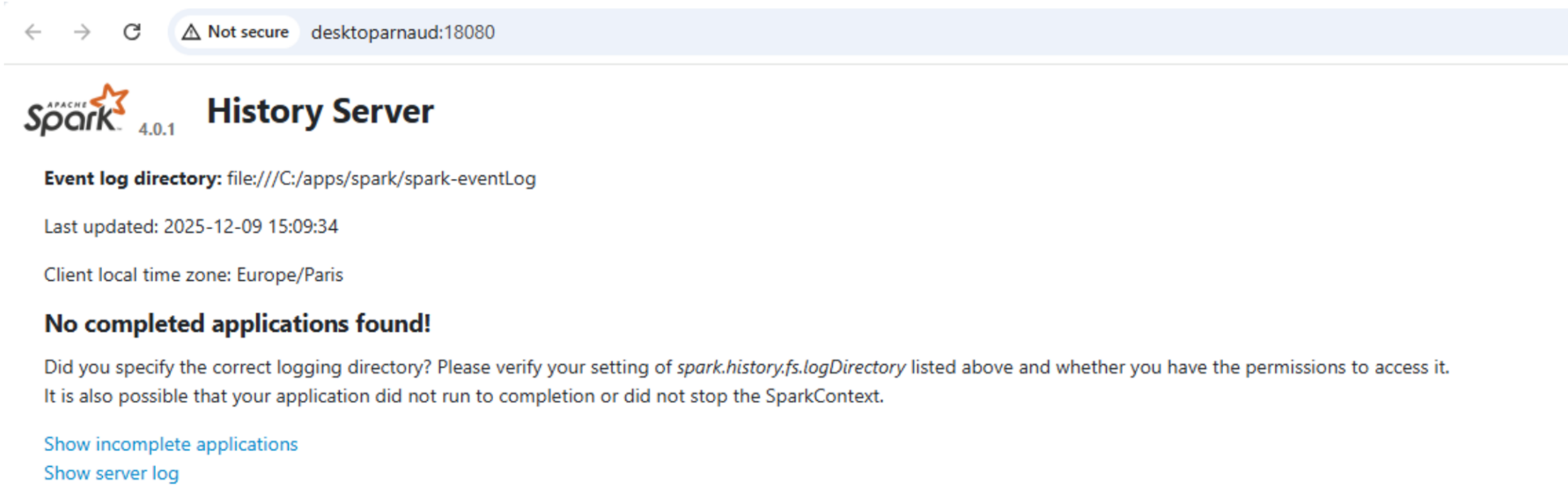
spark.eventLog.dir file:///C:/apps/spark/spark-eventLog

spark.history.fs.logDirectory file:///C:/apps/spark/spark-eventLog

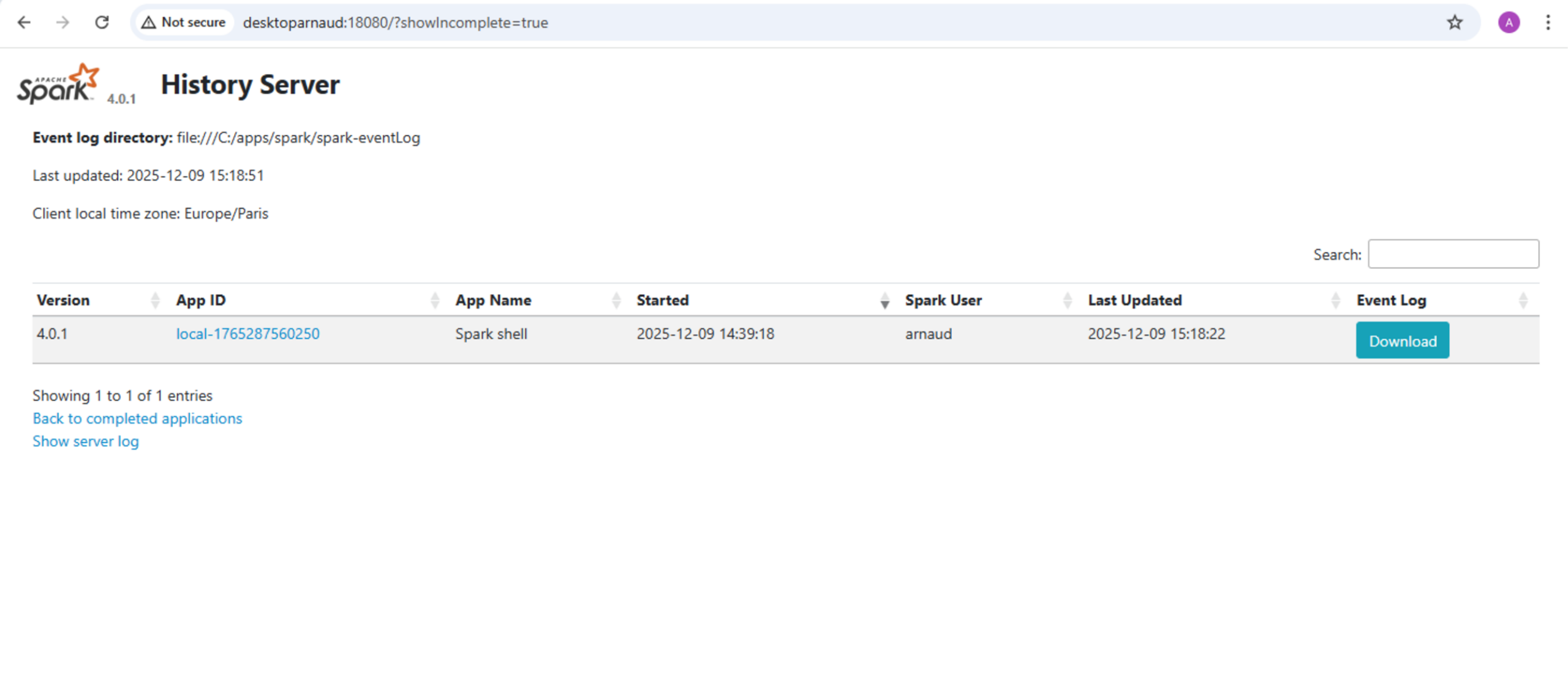
```
25/12/09 15:08:02 INFO Utils: Successfully started service 'HistoryServerUI' on port 18080.
25/12/09 15:08:02 INFO HistoryServer: Bound HistoryServer to 0.0.0.0, and started at http://DesktopArnaud:18080
Exception in thread "main" java.io.FileNotFoundException: Log directory specified does not exist: file:/tmp/spark-events Did you configure
the correct one through spark.history.fs.logDirectory?
        at org.apache.spark.deploy.history.FsHistoryProvider.startPolling(FsHistoryProvider.scala:276)
        at org.apache.spark.deploy.history.FsHistoryProvider.initialize(FsHistoryProvider.scala:221)
```

http://localhost:18080

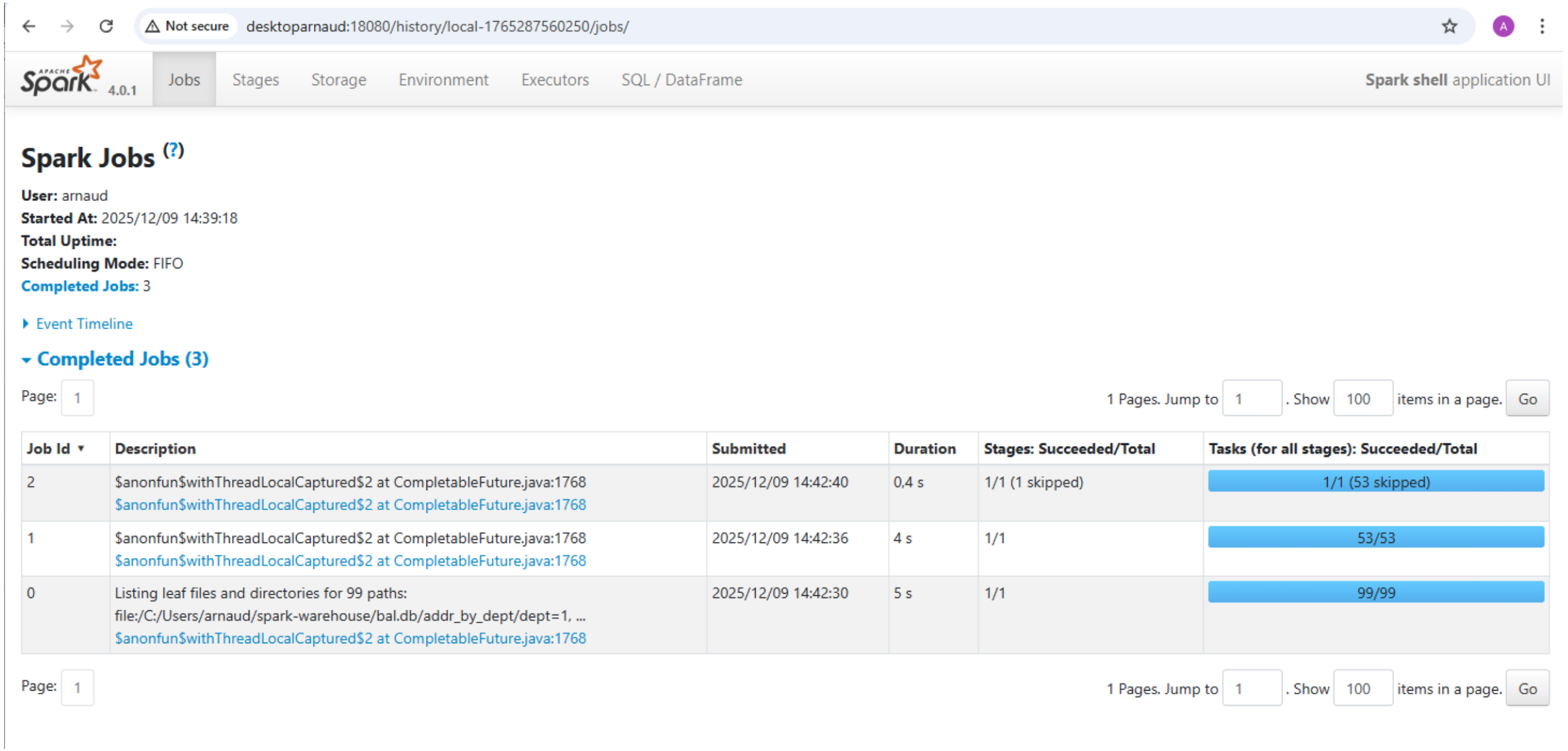
```
25/12/09 15:18:21 INFO JettyUtils: Start Jetty 0.0.0.0:18080 for HistoryServerUI
25/12/09 15:18:21 INFO Utils: Successfully started service 'HistoryServerUI' on port 18080.
25/12/09 15:18:21 INFO HistoryServer: Bound HistoryServer to 0.0.0.0, and started at http://DesktopArnaud:18080
25/12/09 15:18:21 INFO FsHistoryProvider: Parsing file:/C:/apps/spark/spark-eventLog/eventlog_v2_local-1765287560250 for listing data...
25/12/09 15:18:22 INFO FsHistoryProvider: Finished parsing file:/C:/apps/spark/spark-eventLog/eventlog_v2_local-1765287560250
```



Incomplete / Complete Applications

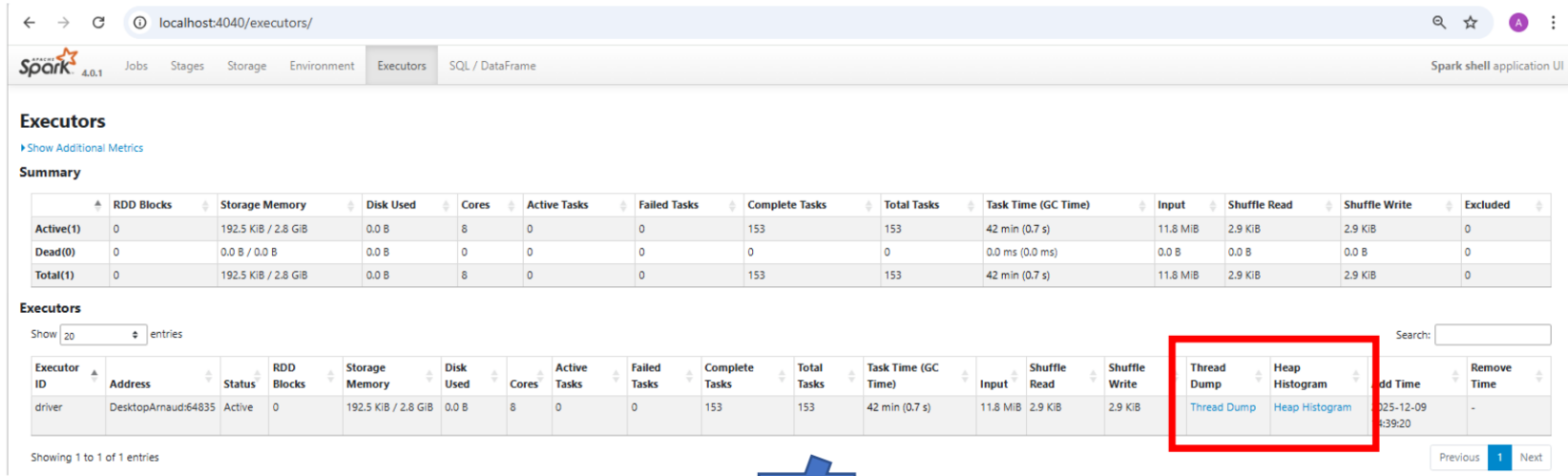


http://localhost:18080/history/${appId}

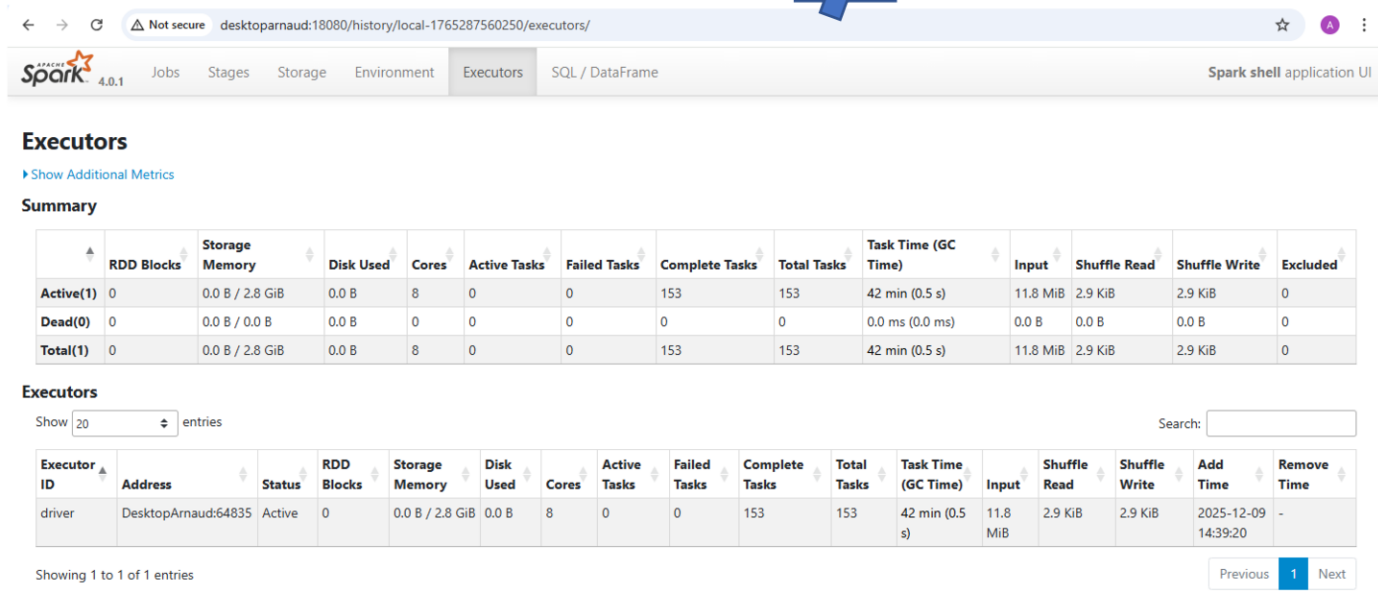


Things You Can NOT Do in Spark History Server … ThreadDumps, Kill Task

Spark UI (live app)



Spark HistoryServer (Offline)

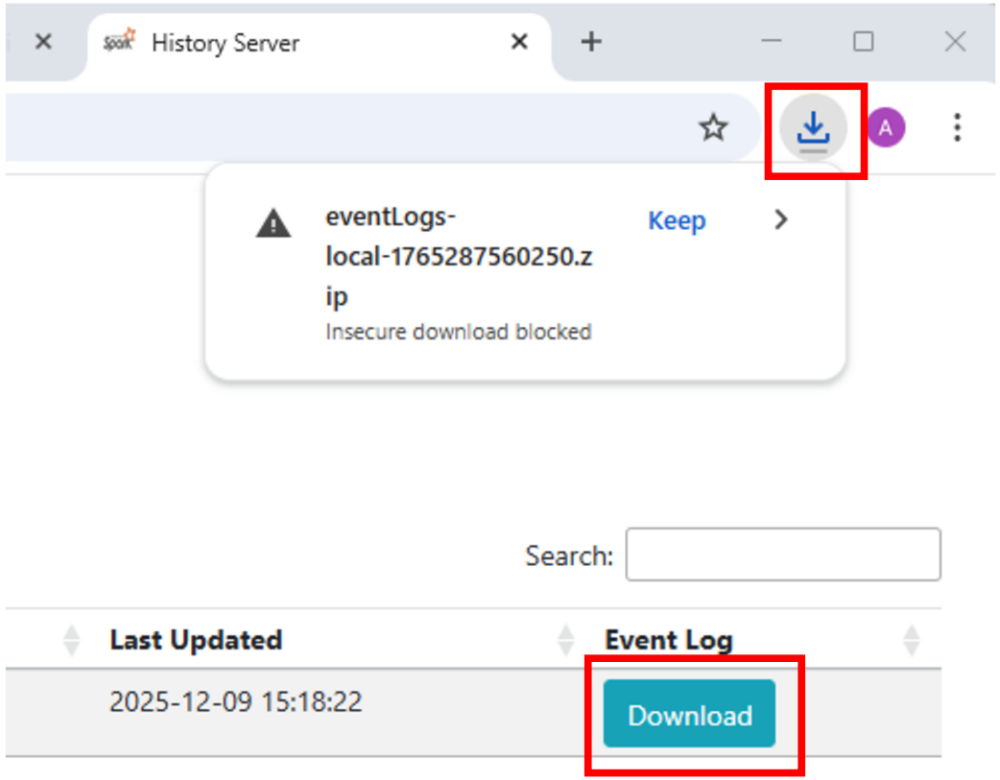


Tip&Tricks for EventLog : Download them

On PROD clusters, eventLogs Files are removed after ~3days
(and may be rewritten, by purging low-level « Task » events)

Download them !!

- for later analyses
- for Requesting Assistance (no credentials / access required)



Spark History Server & Spark UI Limitations

Spark History can do OutOfMemoryError while reloading huge spark eventLog files !!

Both SparkUI & Spark History produce amazingly large & complex web pages
With tons of arrows and blue boxes…

Keep only the last 10 000 spark events, and ignore all others !!

Impossible to Analyse « offline » a very long-running application
(need to manually uncompress + prune spark events …)

Questions ?

... Next Steps ?